# PSYCHOLOGISCHES EXPERIMENTALPRAKTIKUM

## SOSE 2025

Lecturer: Lingyue Chen

TECHNISCHE UNIVERSITÄT DARMSTADT

Centre for Cognitive Science

Sensorimotor Control & Learning

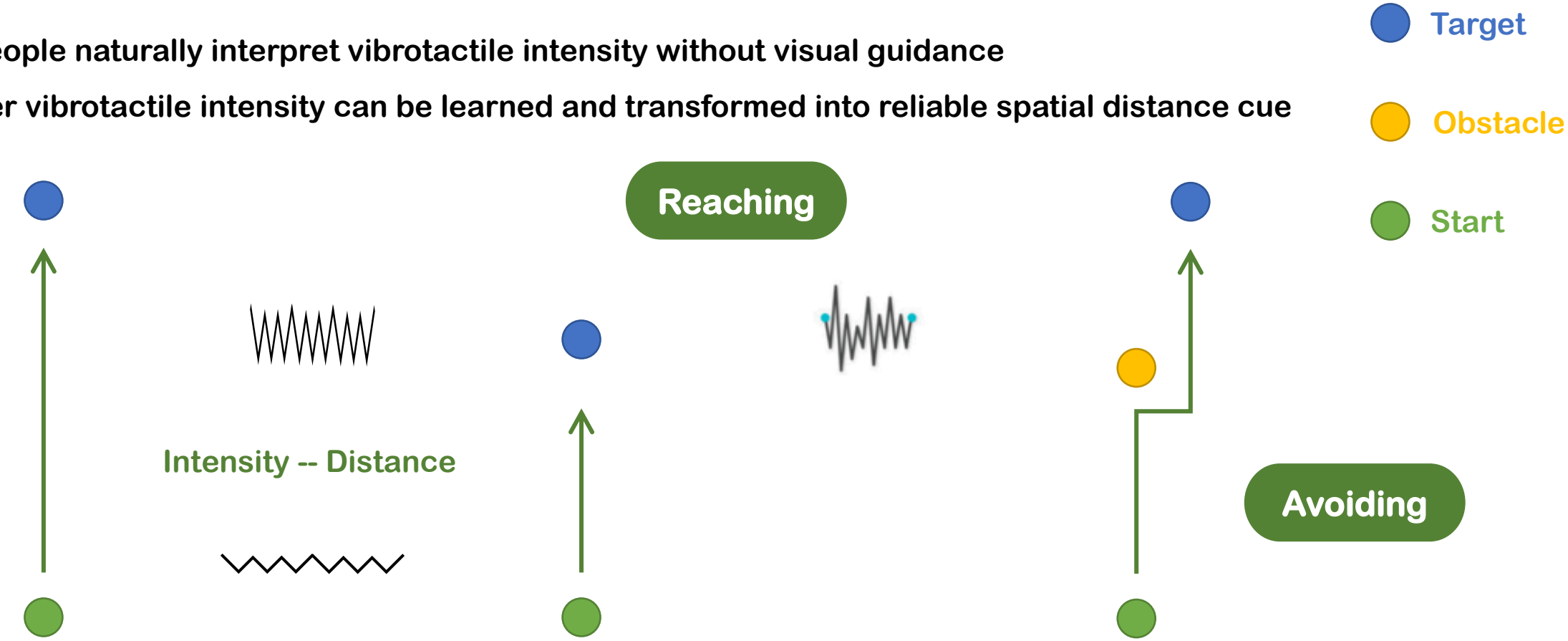
| Week | Date | Topics |
|---|---|---|
| 1 | 24.04 | Project pitches and group assignments |
| 2 | 29.04 | Organization & preparation |
| 3, 4 | 06.05, 13.05 | Research question, study design, grouping |
| 4 – 6 | 14.05 – 30.05 | Experiment coding & preparation |
| 7 – 9 | 02.06 – 20.06 | Data collection |
| 10, 11 | 23.06 – 04.07 | Data analysis & interpretation |
| 12, 13 | 07.07 – 18.07 | Poster preparation |
| 14 | 25.07 | Poster session |
| 19 | 28.08 | Written report submission |

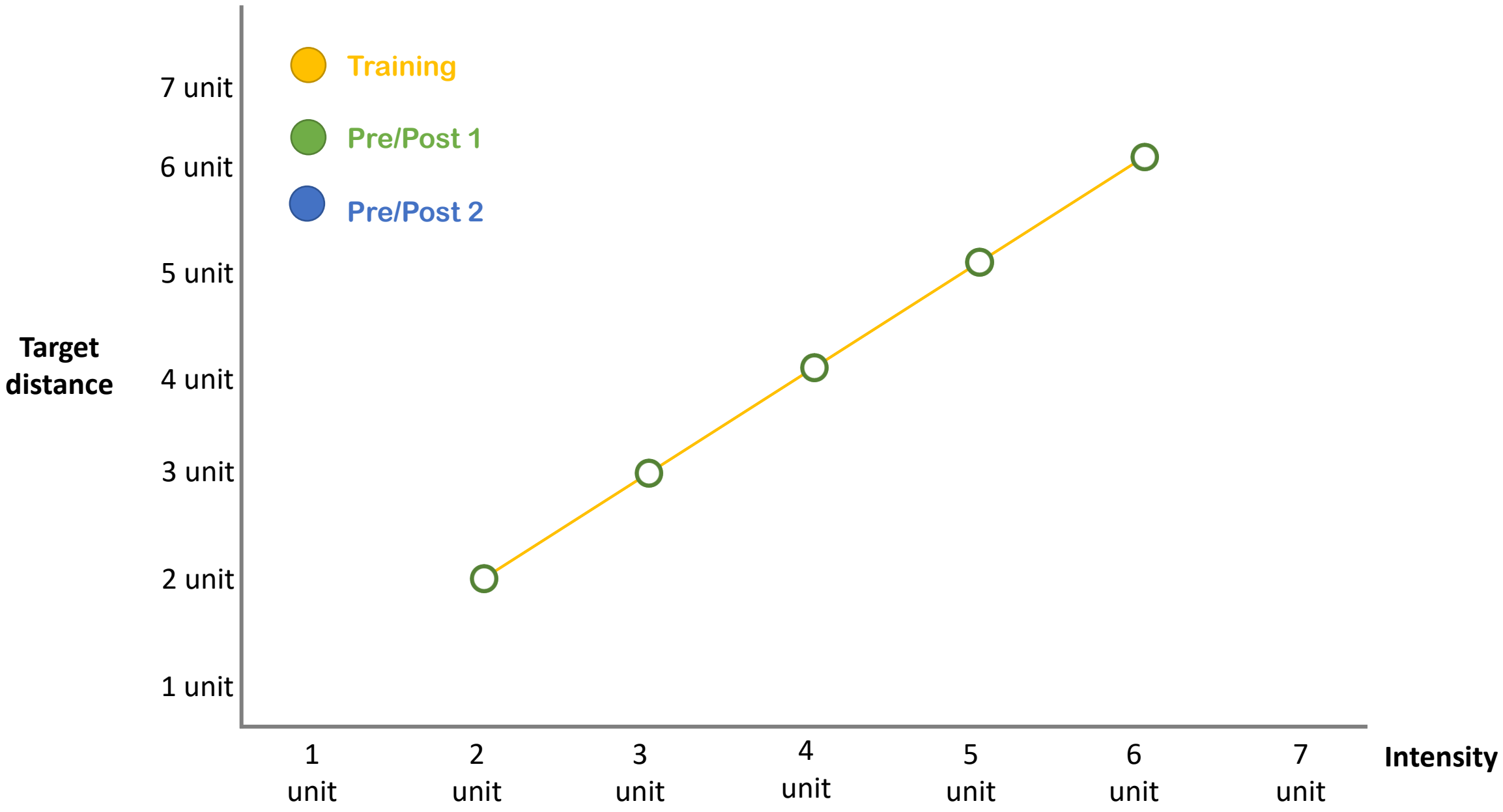
# Task

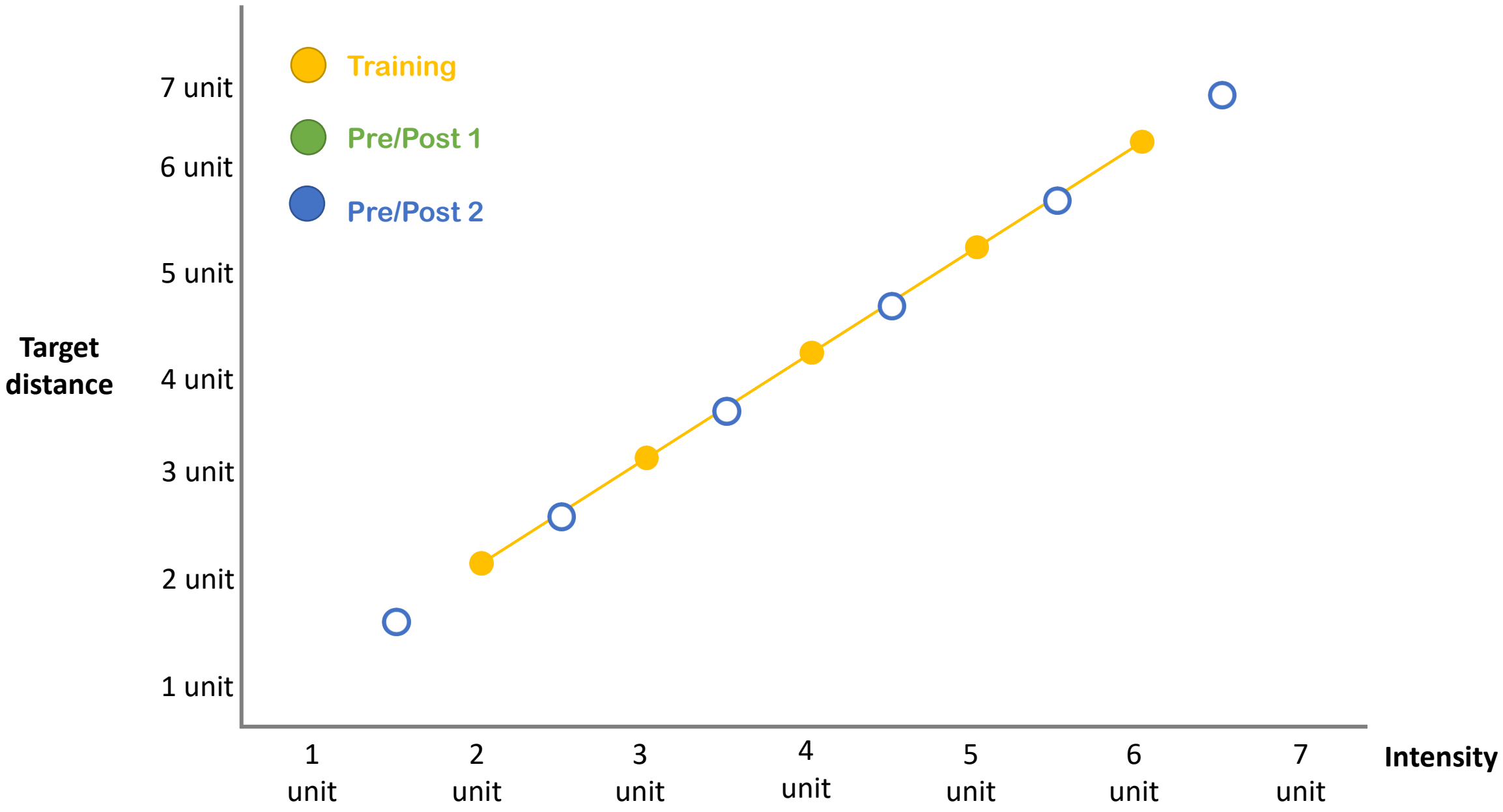
**Sensorimotor mapping**

**Vibrotactile Intensity → Spatial distance**

- How people naturally interpret vibrotactile intensity without visual guidance
- Whether vibrotactile intensity can be learned and transformed into reliable spatial distance cue




Training
Pre/Post 1
Pre/Post 2
Target distance
7 unit
6 unit
5 unit
4 unit
3 unit
2 unit
1 unit
1 unit
2 unit
3 unit
4 unit
5 unit
6 unit
7 unit
Intensity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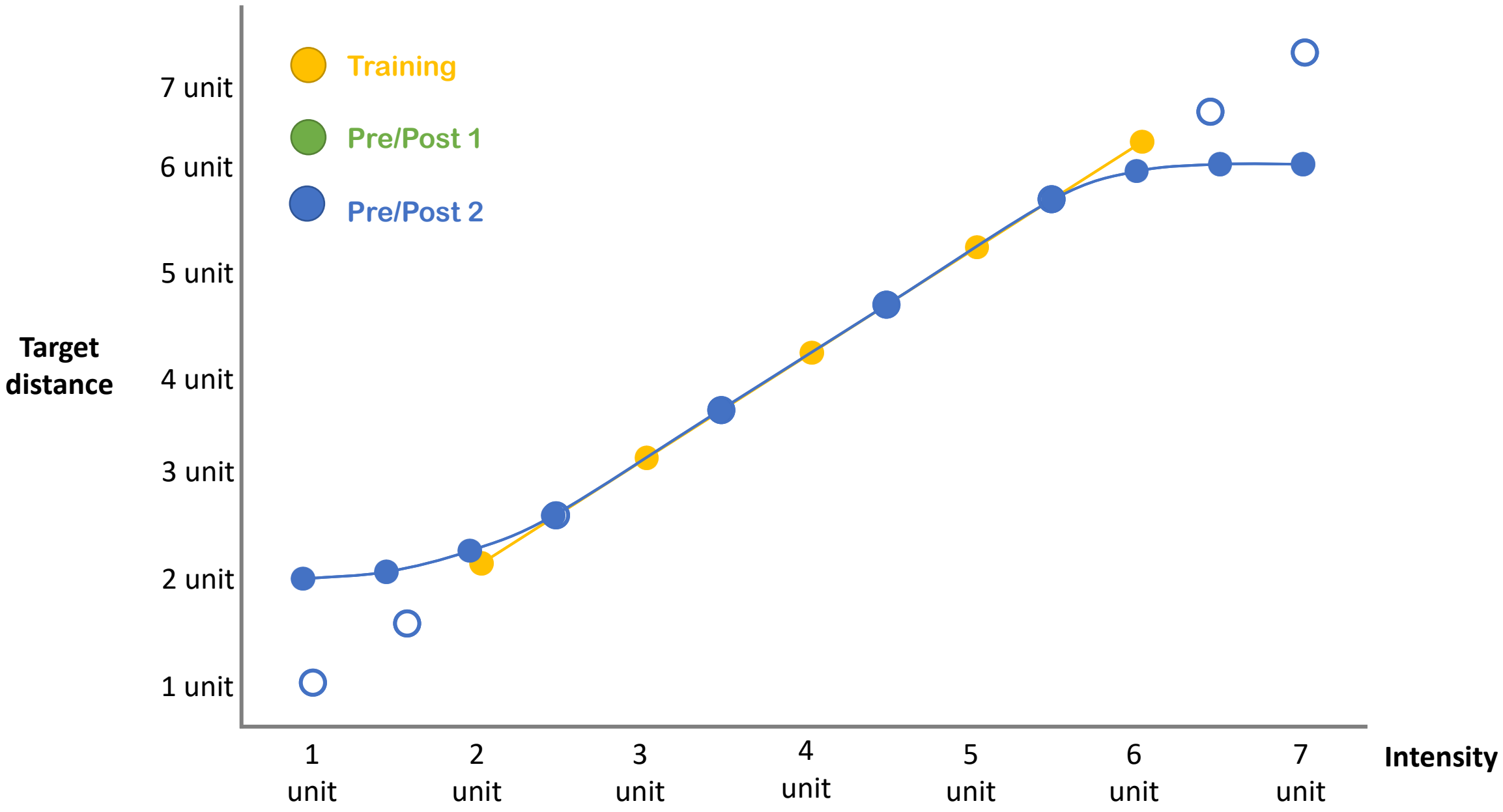
2 unit
3 unit
4 unit
5 unit
6 unit
7 unit
Intensity

# Study Design

| | | | | | | |
|---|---|---|---|---|---|---|
| **Group 1** | Avoiding | Reaching | Reaching | Reaching | Avoiding | Reaching |
| **Group 2** | Reaching | Avoiding | Avoiding | Avoiding | Reaching | Avoiding |
| Block No | 1 | 1 | 5 | 1 | 1 | 1 |
| Target Distance No | 5 + 6 | 5 + 6 | 5 | 5 + 6 | 5 + 6 | 5 + 6 |
| Tactile intensity Level | 5 + 6 | 5 + 6 | 5 | 5 + 6 | 5 + 6 | 5 + 6 |
| Target Repetition No | 8 | 8 | 10 | 8 | 8 | 2 |
| Total Trial No / Block | 88 | 88 | 50 | 88 | 88 | 22 |
| Total Trial No | 176 | | 250 | 176 | | 22 |
| | Pre-test | | Training | Post-test | | Mini-Post |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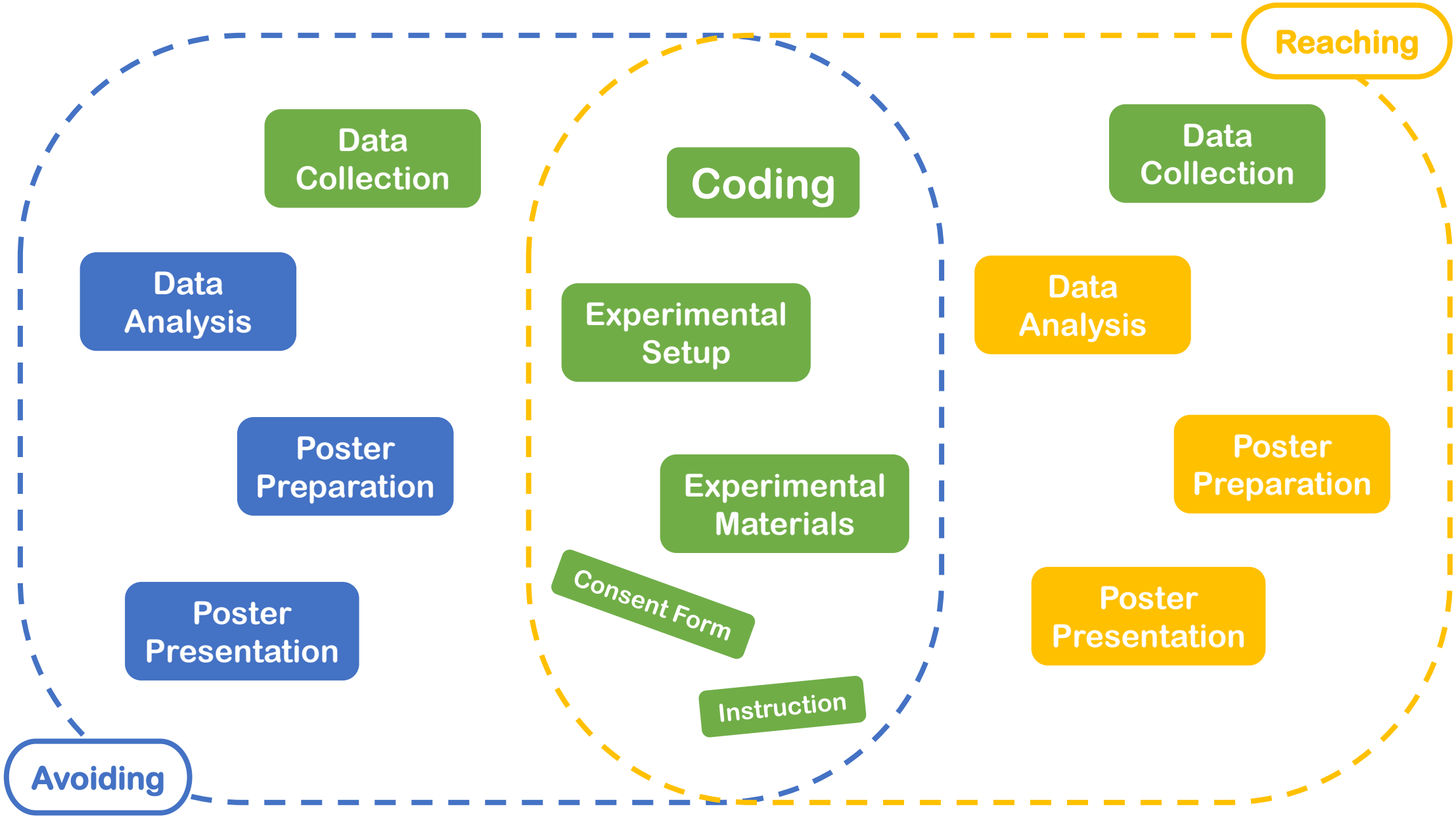
2 unit
3 unit
4 unit
5 unit
6 unit
7 unit
Intensity

# Study Design

| | | | | | | |
|---|---|---|---|---|---|---|
| **Group 1** | Avoiding | Reaching | Reaching | Reaching | Avoiding | Reaching |
| **Group 2** | Reaching | Avoiding | Avoiding | Avoiding | Reaching | Avoiding |
| Block No | 1 | 1 | 5 | 1 | 1 | 1 |
| Target Distance No | 2 + 4 + 5 + 2 | 2 + 4 + 5 + 2 | 5 | 2 + 4 + 5 + 2 | 2 + 4 + 5 + 2 | 2 + 4 + 5 + 2 |
| Tactile intensity Level | 2 + 4 + 5 + 2 | 2 + 4 + 5 + 2 | 4 | 2 + 4 + 5 + 2 | 2 + 4 + 5 + 2 | 2 + 4 + 5 + 2 |
| Target Repetition No | 8 | 8 | 10 | 8 | 8 | 2 |
| Total Trial No / Block | 104 | 104 | 40 | 104 | 104 | 26 |
| Total Trial No | 208 | | 200 | 240 | | 30 |
| | Pre-test | | Training | Post-test | | Mini-Post |


Reaching
Data Collection
Data Analysis
Poster Preparation
Poster Presentation
Coding
Experimental Setup
Experimental Materials
Consent Form
Instruction
Data Collection
Data Analysis
Poster Preparation
Poster Presentation
Avoiding

# Meeting Time

- Group Meeting
  - S1|15 128, Tuesday 13:30 / 15:00
  - Weekly plan
    - **Week 5 – 6: Task group**
    - Week 7 – 9: On request
    - Week 10 & 11: Project group

Experiment

Coding

- Joint Meeting (2 groups)
  - S1|15 015a, Tuesday 13:30
  - Week 12
  - Present your results to the other project group

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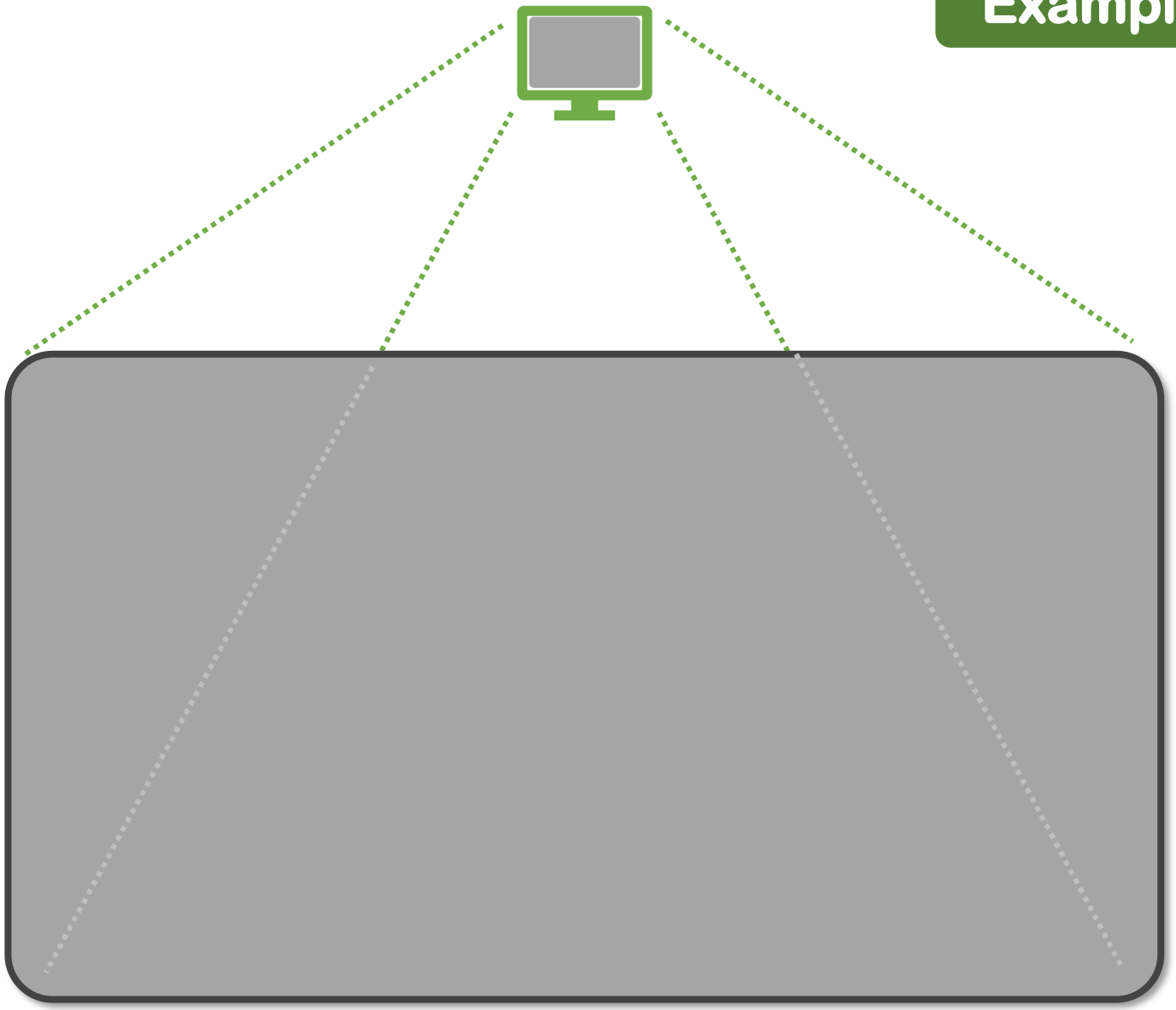
# Example Instructions



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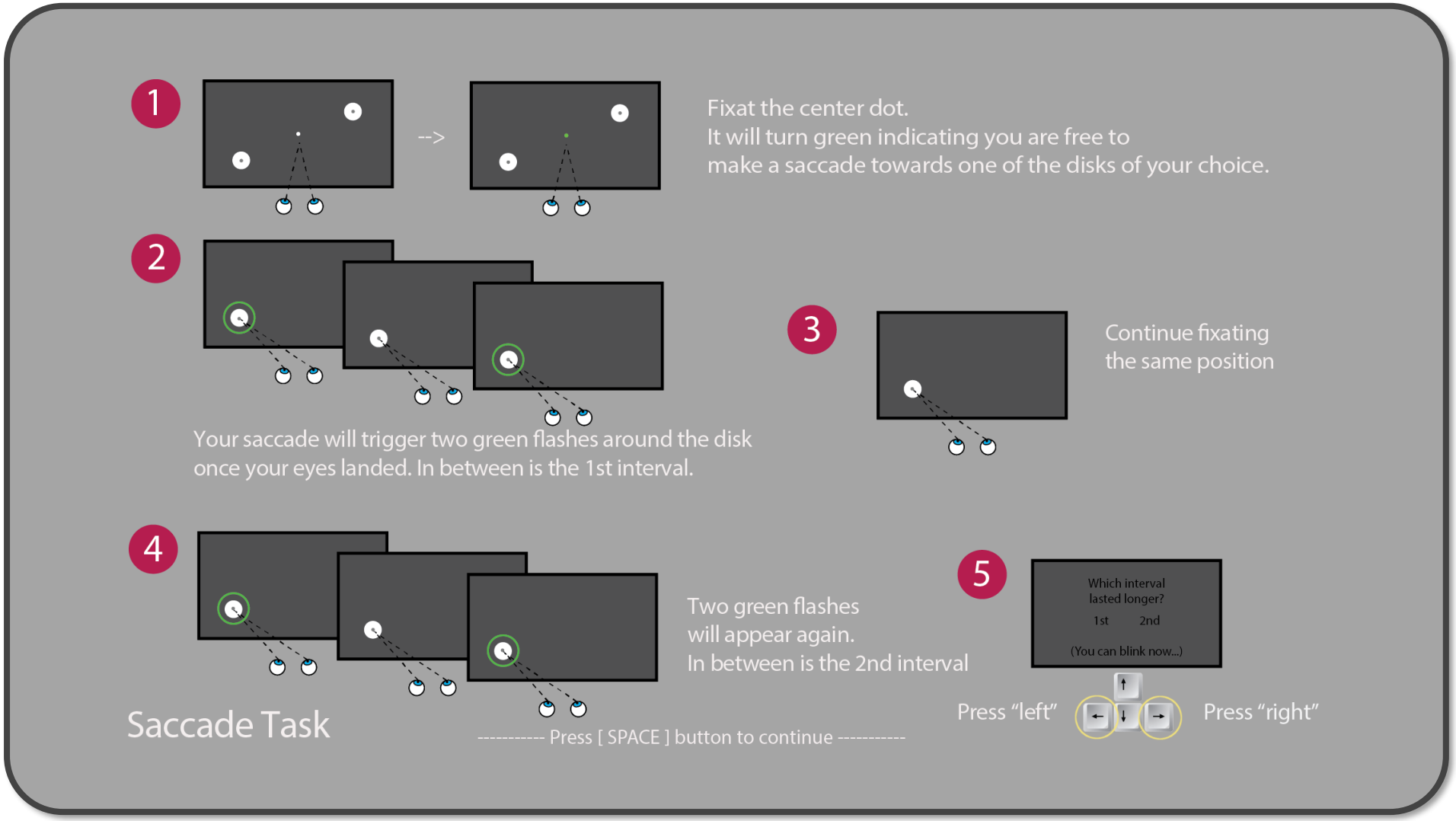
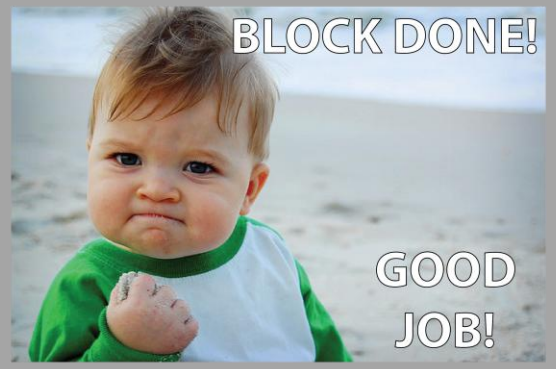


# Example Instructions

Ready for the next block?

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